

The figure displays schematic representations of various genetic constructs. The constructs are organized into two main columns. The left column shows constructs JAC1-3 and MOV1-3, which are fused to a g-g-690V sequence. The right column shows constructs S20K550, GDT1-3, and MOV2-3, which are fused to a g-g-20kA sequence. Each construct is represented by a horizontal line with various colored boxes and labels indicating specific genetic elements.

**Left Column (g-g-690V fusion):**

- JAC1-3:** A green box labeled JAC1-3 is fused to a g-g-690V sequence. The construct is labeled STUDB-M8.
- MOV1-3:** A green box labeled MOV1-3 is fused to a g-g-690V sequence. The construct is labeled S20K550.
- JAC2-3:** A green box labeled JAC2-3 is fused to a g-g-690V sequence. The construct is labeled STUDB-M8.
- MOV2-3:** A green box labeled MOV2-3 is fused to a g-g-690V sequence. The construct is labeled S20K550.
- JAC3-3:** A green box labeled JAC3-3 is fused to a g-g-690V sequence. The construct is labeled STUDB-M8.
- MOV3-3:** A green box labeled MOV3-3 is fused to a g-g-690V sequence. The construct is labeled S20K550.

**Right Column (g-g-20kA fusion):**

- S20K550:** A green box labeled S20K550 is fused to a g-g-20kA sequence. The construct is labeled S20K550.
- GDT1-3:** A green box labeled GDT1-3 is fused to a g-g-20kA sequence. The construct is labeled GDT-3k5-20kA.
- GDT2-3:** A green box labeled GDT2-3 is fused to a g-g-20kA sequence. The construct is labeled GDT-3k5-20kA.
- GDT3-3:** A green box labeled GDT3-3 is fused to a g-g-20kA sequence. The construct is labeled GDT-3k5-20kA.
- MOV2-3:** A green box labeled MOV2-3 is fused to a g-g-20kA sequence. The construct is labeled S20K550.
- MOV3-3:** A green box labeled MOV3-3 is fused to a g-g-20kA sequence. The construct is labeled S20K550.

The constructs are labeled with their respective names (JAC1-3, MOV1-3, JAC2-3, MOV2-3, JAC3-3, MOV3-3, S20K550, GDT1-3, GDT2-3, GDT3-3) and the fusion sequence (g-g-690V or g-g-20kA). The constructs are also labeled with their respective names (STUDB-M8, S20K550, GDT-3k5-20kA) and the fusion sequence (g-g-690V or g-g-20kA).

Figure 1: Schematic diagrams of the CMC-3PH-2mH-SKU and CMC-3PH-2mH-SKU circuit topologies. The top row shows the LOM1, CX11, and CX12 components. The middle row shows the CY1 component. The bottom row shows the LOM2, CX13, CX23, and CY3 components. The CMC-3PH-2mH-SKU is shown on the left, and the CMC-3PH-2mH-SKU is shown on the right. The diagrams illustrate the internal structure of each component, including the LOM (Low Order Modulator) and CX (Cross-Modulator) blocks, and their interconnections. The LOM1 and LOM2 blocks are shown with their internal components and connections. The CX11, CX12, CX13, and CX23 blocks are shown with their internal components and connections. The CY1 and CY3 blocks are shown with their internal components and connections. The diagrams are labeled with component names and values, such as 20μH, 2.2μF, and 4.7μF.

Figure 1 shows the schematic diagram of the HF167F-80A-M and CER-25W-AX components. The HF167F-80A-M is a 16-pin DIP package. It features a KP1E1 block (pins 1-4), a KP1E2 block (pins 5-8), and a RELAY\_AIR\_4076 block (pins 9-16). The CER-25W-AX is a 4-pin package. It features a KP1E1 block (pins 1-2) and a RELAY\_AIR\_4076 block (pins 3-4). The diagram illustrates the internal connections between these blocks and the external pins.

Figure 1: Schematic diagrams of the 120V AC power supply circuit. The figure shows 12 sub-circuits labeled (A) through (L), each representing a different combination of components (CT, DI, DI2, DI3, DI4, DI5, DI6, DI7, DI8, DI9, DI10, DI11, DI12) and their connections to the AC supply and the ACX-1100 unit. The components are represented by colored boxes: yellow for CT, orange for DI, and green for DI2, DI3, DI4, DI5, DI6, DI7, DI8, DI9, DI10, DI11, DI12. The connections are shown with lines and labels indicating the component type and value.

Figure 1 shows 12 different configurations of the 1200 V SiC MOSFETs, labeled CD704 through CD715. Each configuration is represented by a schematic diagram showing the internal structure and key parameters:

- CD704:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD705:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD706:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD707:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD708:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD709:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD710:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD711:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD712:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD713:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD714:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.
- CD715:** Features a  $SiC_{0.9999999999999999}$  gate stack, a  $SiC_{0.9999999999999999}$  channel, and a  $SiC_{0.9999999999999999}$  drift region. The gate length is 470 nm, and the channel length is 470 nm.

Figure 1 shows schematic diagrams of the four transgenic lines. (A) CER-25W-AX: A construct with a 5' UTR, RDH50, and a 3' UTR. (B) CER-25W-AX: A construct with a 5' UTR, RDH51, and a 3' UTR. (C) CER-25W-AX: A construct with a 5' UTR, RDH52, and a 3' UTR. (D) CER-25W-AX: A construct with a 5' UTR, RDH53, and a 3' UTR. (E) SIC-1200-5A: A construct with a 5' UTR, CDSF, and a 3' UTR. (F) TAB-M4: A construct with a 5' UTR, TAB, and a 3' UTR. (G) TLP152-class: A construct with a 5' UTR, RDGL, and a 3' UTR. (H) QA01C: A construct with a 5' UTR, RDGL, and a 3' UTR.

[illegible][illegible]

ULN2803A

Pin 1: CT1, XSP5  
 Pin 2: IN1  
 Pin 3: IN2  
 Pin 4: IN3  
 Pin 5: IN4  
 Pin 6: IN5  
 Pin 7: IN6  
 Pin 8: IN7  
 Pin 9: IN8  
 Pin 10: COM  
 Pin 11: NC, OMR  
 Pin 12: NC, O18  
 Pin 13: NC, O6  
 Pin 14: NC, O5  
 Pin 15: NC, O4  
 Pin 16: NC, O3  
 Pin 17: NC, O2  
 Pin 18: COB, XSP5

Ground connections: DGND, V24

PH-4-FAN

PH-4-FAN2

[illegible]

30 kW module · AC-DC (lower) board · sheet 1 of 3 · 1x Vienna PFC cell + 1x 3-ph LLC cell  
15 sections · 153 wired clusters · 271 symbols · cross-section links are net labels

generated 2026-09-08  
LCSC: calculations/out/bom-30kw.csv

[illegible]

Figure 1: Schematic representation of the 12 different HFE82V-M-CLASS and SOP-10R-25W gene constructs. The figure is organized into two main sections: HFE82V-M-CLASS (left) and SOP-10R-25W (right). Each section contains three rows of constructs. The top row shows constructs with a V24 promoter, a 5' UTR, a CDS (containing a stop codon), and a 3' UTR. The middle row shows constructs with a V24 promoter, a 5' UTR, a CDS (containing a stop codon), and a 3' UTR. The bottom row shows constructs with a V24 promoter, a 5' UTR, a CDS (containing a stop codon), and a 3' UTR. The constructs are labeled as follows: HFE82V-M-CLASS (top row: V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER; middle row: V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER; bottom row: V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER); SOP-10R-25W (top row: V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER; middle row: V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER; bottom row: V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER, V24, RPUSKPER).

Figure 1: Schematic diagrams of the four CRISPR-Cas9 systems used in the study. The diagrams illustrate the genomic organization of the CRISPR-Cas9 systems, including the Cas9 genes (A, B, S) and the CRISPR arrays (R) flanked by inverted repeats (IR). The systems are labeled as follows:

- (A) VOM1271T system: RPVLA, LVPVA, and RPVBA.
- (B) SIC-1200-5A system: RPVLB, LVPB, and RPVB.
- (C) CER-2k2-10W-AX system: RPVLA, LVPVA, and RPVBA.
- (D) SIC-1200-5A system: RPVLB, LVPB, and RPVB.

[illegible]

The schematic diagram illustrates the internal components of the LED-2DIG-0.56CC module. At the top, a 74HC95 counter is connected to a 5V supply (V5) and ground (GND). The counter's outputs (Q0-Q4) are connected to the RS485 transceiver (RS485) and the LED driver (LED-2DIG-0.56CC). The RS485 transceiver is connected to a 485+ signal line and a 485- signal line. The LED driver is connected to a 5V supply and ground. The module features two 7-segment displays (DISP1 and DISP2) and two push buttons (SW1 and SW2). The module is labeled 'LED-2DIG-0.56CC'.

**V2P3**

1000

100

RBFLVX

LINC\_205

LINC\_18

**V24**

1000

100

RBFLVX

LINC\_18

LINC\_19

**V2P37**

1000

100

RBFLVX

LINC\_18

LINC\_19

The diagram illustrates the genetic constructs used in the study. It shows three constructs: V2P3, V24, and V2P37. Each construct is represented by a horizontal line with a scale bar indicating 1000 bp and a 100 bp segment. V2P3 contains the RBFLVX gene (green box) flanked by LINC\_205 (blue box) and LINC\_18 (blue box). V24 contains the RBFLVX gene flanked by LINC\_18 and LINC\_19 (blue box). V2P37 contains the RBFLVX gene flanked by LINC\_18 and LINC\_19. The RBFLVX gene is shown as a green box with a red border. The LINC regions are shown as blue boxes. The RBFLVX gene is flanked by LINC\_205 and LINC\_18 in V2P3, LINC\_18 and LINC\_19 in V24, and LINC\_18 and LINC\_19 in V2P37. The RBFLVX gene is flanked by LINC\_205 and LINC\_18 in V2P3, LINC\_18 and LINC\_19 in V24, and LINC\_18 and LINC\_19 in V2P37. The RBFLVX gene is flanked by LINC\_205 and LINC\_18 in V2P3, LINC\_18 and LINC\_19 in V24, and LINC\_18 and LINC\_19 in V2P37.

[illegible]

Pin connection diagram for the ULN2803A. The chip is shown with pins 1 through 16. Pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, and 16 are connected to various components. Pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, and 16 are connected to various components. Pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, and 16 are connected to various components.

[illegible]

30 kW module · DC-DC (upper) board · sheet 2 of 3 · 1x Vienna PFC cell + 1x 3-ph LLC cell  
14 sections · 158 wired clusters · 282 symbols · cross-section links are net labels

generated 2026-09-08  
LCSC: calculations/out/bom-30kw.csv

The diagram illustrates the pin configuration for the GD32G553VET6 microcontroller. The central component is the microcontroller pin header, with pins numbered 1 to 51. The connections are as follows:

- V3P3 Connections:** Pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, and 51 are connected to V3P3.
- DGND Connections:** Pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, and 51 are connected to DGND.
- Peripheral Connections:**
  - AIN6, AIN7, PWM6, PWM5, PWM4, PWM2:** Connected to pins 15, 16, 17, 18, 19, and 20 respectively.
  - RCARDRST:** Connected to pin 21.
  - NRST\_CARD:** Connected to pin 22.
  - CCARDD2:** Connected to pin 23.
  - CCARDD1:** Connected to pin 24.
  - CCARDD3:** Connected to pin 25.
  - CCARDD4:** Connected to pin 26.
  - CCARDA1:** Connected to pin 27.
  - CCARDA2:** Connected to pin 28.
  - CCARDDVR:** Connected to pin 29.
  - FBCARDA:** Connected to pin 30.
  - VDDA\_CARD:** Connected to pin 31.

The diagram also shows various capacitors and resistors used for decoupling and signal conditioning, such as 100nF capacitors for V3P3 and DGND, and 10kΩ resistors for NRST\_CARD and CCARDD2.

**CAN-CARD-88**

| Pin | Signal | Pin | Signal  |
|-----|--------|-----|---------|
| 1   | PWM0   | 88  | PWM0    |
| 2   | PWM1   | 87  | PWM1    |
| 3   | PWM2   | 86  | PWM2    |
| 4   | PWM3   | 85  | PWM3    |
| 5   | PWM4   | 84  | PWM4    |
| 6   | PWM5   | 83  | PWM5    |
| 7   | PWM6   | 82  | PWM6    |
| 8   | PWM7   | 81  | PWM7    |
| 9   | PWM8   | 80  | PWM8    |
| 10  | PWM9   | 79  | PWM9    |
| 11  | PWM10  | 78  | PWM10   |
| 12  | PWM11  | 77  | PWM11   |
| 13  | AIN0   | 76  | AIN0    |
| 14  | AIN1   | 75  | AIN1    |
| 15  | AIN2   | 74  | AIN2    |
| 16  | AIN3   | 73  | AIN3    |
| 17  | AIN4   | 72  | AIN4    |
| 18  | AIN5   | 71  | AIN5    |
| 19  | AIN6   | 70  | AIN6    |
| 20  | AIN7   | 69  | AIN7    |
| 21  | AIN8   | 68  | AIN8    |
| 22  | AIN9   | 67  | AIN9    |
| 23  | AIN10  | 66  | AIN10   |
| 24  | AIN11  | 65  | AIN11   |
| 25  | AIN12  | 64  | AIN12   |
| 26  | TSNS0  | 63  | TSNS0   |
| 27  | TSNS1  | 62  | TSNS1   |
| 28  | AVMD   | 61  | AVMD    |
| 29  | AGND_2 | 60  | AGND_2  |
| 30  | DO0    | 59  | DO0     |
| 31  | DO1    | 58  | DO1     |
| 32  | DO2    | 57  | DO2     |
| 33  | DO3    | 56  | DO3     |
| 34  | DO4    | 55  | DO4     |
| 35  | DO5    | 54  | DO5     |
| 36  | DO6    | 53  | DO6     |
| 37  | DIO3   | 52  | DIO3    |
| 38  | DIO4   | 51  | DIO4    |
| 39  | DIO5   | 50  | DIO5    |
| 40  | DIO6   | 49  | DIO6    |
| 41  | DIO7   | 48  | DIO7    |
| 42  | DIO8   | 47  | DIO8    |
| 43  | DIO9   | 46  | DIO9    |
| 44  | DIO10  | 45  | DIO10   |
|     |        | 44  | EN_A    |
|     |        | 43  | EN_B    |
|     |        | 42  | DRV_RDY |
|     |        | 41  | LINK_TX |
|     |        | 40  | LINK_RX |
|     |        | 39  | CAN_TX  |
|     |        | 38  | CAN_RX  |
|     |        | 37  | HM10    |
|     |        | 36  | HM11    |
|     |        | 35  | HM12    |
|     |        | 34  | HM13    |
|     |        | 33  | HM14    |
|     |        | 32  | HM15    |
|     |        | 31  | HM16    |
|     |        | 30  | ROLE0   |
|     |        | 29  | ROLE1   |
|     |        | 28  | V15     |
|     |        | 27  | V3P3    |
|     |        | 26  | GND     |
|     |        | 25  | AGND    |
|     |        | 24  | AGND    |
|     |        | 23  | AGND    |
|     |        | 22  | AGND    |
|     |        | 21  | AGND    |
|     |        | 20  | AGND    |
|     |        | 19  | AGND    |
|     |        | 18  | AGND    |
|     |        | 17  | AGND    |
|     |        | 16  | AGND    |
|     |        | 15  | AGND    |
|     |        | 14  | AGND    |
|     |        | 13  | AGND    |
|     |        | 12  | AGND    |
|     |        | 11  | AGND    |
|     |        | 10  | AGND    |
|     |        | 9   | AGND    |
|     |        | 8   | AGND    |
|     |        | 7   | AGND    |
|     |        | 6   | AGND    |
|     |        | 5   | AGND    |
|     |        | 4   | AGND    |
|     |        | 3   | AGND    |
|     |        | 2   | AGND    |
|     |        | 1   | AGND    |

**CONN-CARD-88**

The schematic diagram illustrates the electrical connections for the Uuand card. It features two main integrated circuits: a 74HC11 and a TPS3430-class.

**74HC11 Connections:**

- Inputs:** EN A (pin 1) and EN B (pin 2) are connected to the input pins. A pull-down resistor is connected from EN A to DGND. A pull-up resistor is connected from EN B to V3P3.
- Outputs:** Y1 (pin 12) is connected to the WDO\_CARD pin. Y2 (pin 9) is connected to the GND pin.
- Power:** VCC (pin 14) is connected to V3P3. GND (pin 7) is connected to DGND.

**TPS3430-class Connections:**

- Inputs:** WDI (pin 1) is connected to the input pin. A pull-down resistor is connected from WDI to DGND. GND (pin 2) is connected to DGND. SET0 (pin 3) and SET1 (pin 4) are connected to the input pins.
- Outputs:** CRST (pin 8) is connected to the CRST\_CARD pin. A pull-up resistor is connected from CRST\_CARD to V3P3. WDO (pin 5) is connected to the WDO\_CARD pin.
- Power:** VDD (pin 6) is connected to V3P3. GND (pin 7) is connected to DGND.

**Other Connections:**

- Resistors:** A 100kΩ resistor is connected from EN B to V3P3. A 100kΩ resistor is connected from WDO\_CARD to V3P3. A 10kΩ resistor is connected from WDO\_CARD to DGND. A 10kΩ resistor is connected from RRDY\_CARD to V3P3. A 10kΩ resistor is connected from RGPDCARD to V3P3.
- Capacitors:** A 100nF capacitor is connected from V3P3 to CSFCARD. A 1nF capacitor is connected from V3P3 to CWD CARD. A 1nF capacitor is connected from V3P3 to CRST\_CARD.

The schematic diagram illustrates the power supply circuit for the TPS54202-class converter. It features three main components: UBKCARD, CBKCARD, and CBKCARD.

- UBKCARD:** A yellow rectangular component with pins 1 (VIN), 2 (GND), 3 (SW), 4 (FB), 5 (EN), 6 (BST), 7 (NC1), and 8 (NC2). It is connected to a 100nF capacitor (labeled 100nF) and a 10μH inductor (labeled 10μH).
- CBKCARD:** A green rectangular component with pins 1 (VIN), 2 (GND), 3 (SW), 4 (FB), 5 (EN), 6 (BST), 7 (NC1), and 8 (NC2). It is connected to a 100nF capacitor (labeled 100nF) and a 10μH inductor (labeled 10μH).
- CBKCARD:** A green rectangular component with pins 1 (VIN), 2 (GND), 3 (SW), 4 (FB), 5 (EN), 6 (BST), 7 (NC1), and 8 (NC2). It is connected to a 100nF capacitor (labeled 100nF) and a 10μH inductor (labeled 10μH).

The circuit is powered by a 15V input (V15) and includes a 100nF capacitor (labeled 100nF) and a 10μH inductor (labeled 10μH). The output is connected to a 100nF capacitor (labeled 100nF) and a 10μH inductor (labeled 10μH).

[illegible]

|                                                                                                                                                                                              |                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>DC-Modules — Control Card (GD32G553VET6, one card for both converter roles)</b>                                                                                                           | <b>Rev D.1</b>                                                      |
| 30/60 kW module · Control card board · sheet 3 of 3 · role-agnostic: AC-DC or DC-DC slot via ROLE straps<br>8 sections · 37 wired clusters · 45 symbols · cross-section links are net labels | generated 2026-09-08<br>LCSC: calculations/out/bom-control-card.csv |